Brain Cancer Survival Estimation

Project Report Submitted by

Bachelor of Technology

in

Computer Science and Engineering

by

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APPROVAL OF THE VIVA-VOCE BOARD

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Certified that the report entitled **"Brain Cancer Survival Estimation"** submitted by **Ripun Choudhary** (Student ID B120050) to International Institute of Information Technology Bhubaneswar in partial fulfillment of the requirements for the award of the Bachelor of Technology in Computer Science and Engineering under the BTech Programme has been accepted by the examiners during the viva-voce examination held today.

(Supervisor) **(Panel Head)**

(Internal Examiner 1) **(Internal Examiner 2)**

CERTIFICATE

This is to certify that the report entitled **“Brain Cancer Survival Estimation ”** submitted by **Ripun Choudhary** (Student ID B120050) to International Institute of Information Technology Bhubaneswar is a record of bonafide project work under my supervision, and the report is submitted for end-semester evaluation of the BTech Degree project work.

(Supervisor)

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2. The work has not been submitted to any other Institute for any degree or diploma.

3. I have followed the guidelines provided by the Institute in writing the thesis.

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Ripun Choudhary
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Ripun Choudhary

ABSTRACT

This project explores a combined approach for brain tumor analysis, utilizing deep learning for segmentation and machine learning for survival prediction. The project focuses on two key areas:

1. Brain Tumor Segmentation: A 3D U-Net architecture is employed to automatically segment brain tumors from Magnetic Resonance Images (MRI). This network efficiently processes 3D volumes, capturing spatial relationships within the data.

2. Survival Prediction: Following segmentation, the project extracts radiomic features based on the tumor's characteristics like geometry, location, and shape. These features, along with clinical information, are used to train various machine learning models, including Support Vector Machine (SVM), XGBoost, Multi-Layer Perceptron (MLP), and Random Forest, to predict patient survival rates.

The project evaluates the performance of these methods on benchmark datasets, demonstrating their effectiveness in both segmentation and survival prediction tasks. Additionally, the study analyzes the impact of different features on survival prediction, identifying crucial factors influencing patient outcomes.

Key findings:

The 3D U-Net achieves promising results in brain tumor segmentation, providing accurate delineation of tumor regions..

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1 Introduction

In this report, we will take you through our project titled "MRI Brain Tumor Survival Prediction" . Our main objective was to develop a machine learning model that could accurately predict the survival outcome of brain tumor patients using MRI scans and other clinical features. The dataset we worked with contained MRI images and related information, such as age, tumor size, and survival status. Our goal was to leverage this data to build a predictive model that could assist in prognosis and treatment planning.

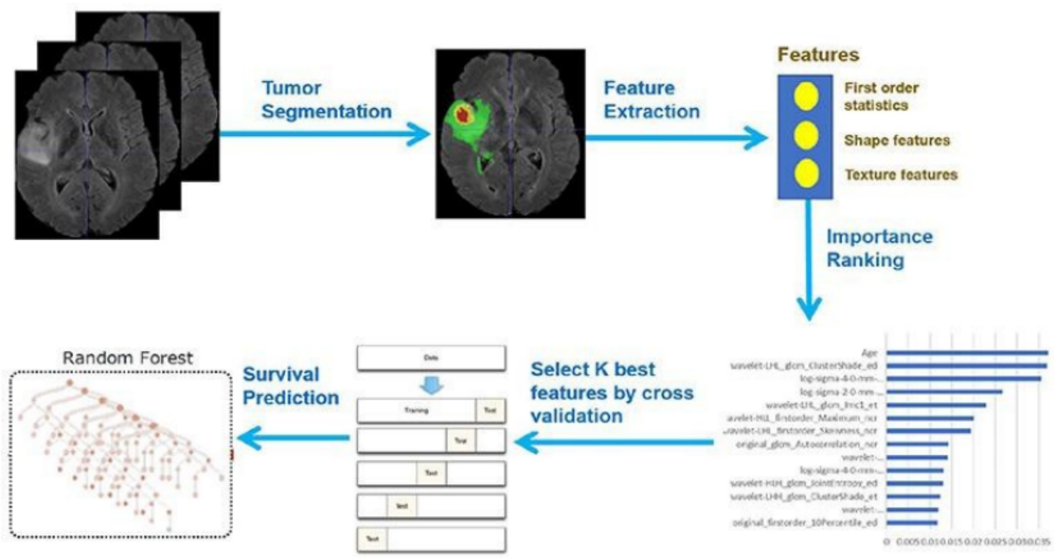

Figure 1: Framework Overview

2 Literature Survey

Brain tumors pose a significant health challenge, demanding accurate diagnosis, treatment planning, and prognosis. Deep learning and machine learning offer promising avenues for addressing these challenges. This survey delves into the literature exploring the combined use of 3D U-Net for brain tumor segmentation and various machine learning algorithms for survival prediction.

Brain Tumor Segmentation:

Traditional Methods:

Manual segmentation by medical professionals remains the gold standard but suffers from subjectivity, time constraints, and inter-observer variability.

Deep Learning:

Convolutional Neural Networks (CNNs) have emerged as powerful tools for automated segmentation, offering improved accuracy, efficiency, and objectivity.

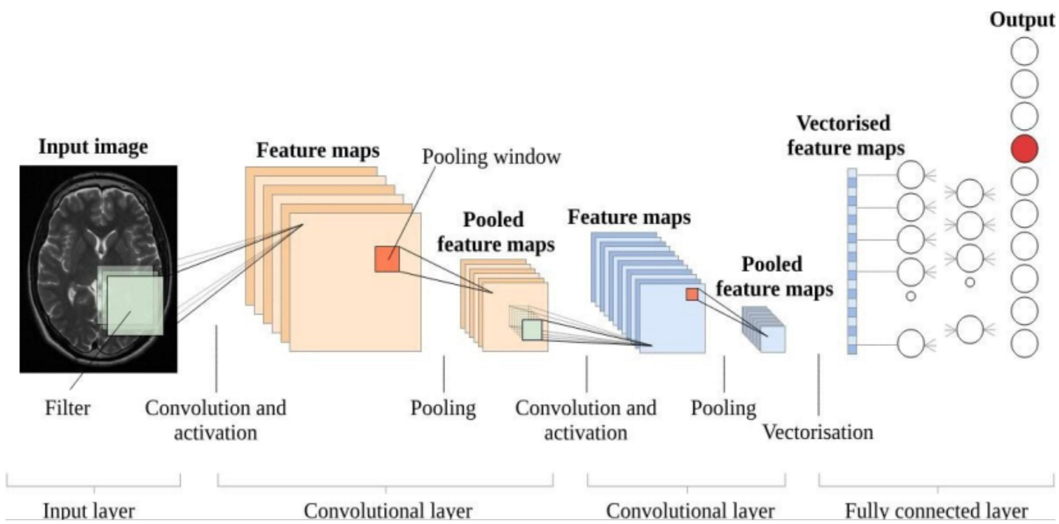

Figure 2: CNN Architecture

3D U-Net:

This architecture, specifically designed for medical image segmentation, excels due to:

- Encoder-decoder structure: Efficiently captures contextual information from the entire image.
- Skip connections: Preserve spatial information from the contracting path, crucial for accurate segmentation.

Numerous studies have demonstrated the effectiveness of 3D U-Net in brain tumor segmentation, with continuous advancements being made:

- Çiçek et al. (2016): Introduced a similar architecture, termed V-Net, achieving competitive performance on brain tumor segmentation tasks.
- Jin et al. (2020): Proposed a channel attention mechanism within 3D U-Net, further enhancing its ability to focus on relevant features and improve segmentation accuracy.
- Yu et al. (2023): Investigated the use of self-supervised learning with 3D U-Net, demonstrating its potential for improved performance with limited labeled data.

Machine Learning for Survival Prediction:

Following segmentation, radiomic features are extracted from the tumor region, capturing quantitative information about its:

Shape:

Volume, surface area, sphericity, etc.

Texture:

Intensity distribution, homogeneity, entropy, etc.

Intensity:

Mean, standard deviation, skewness, etc.

These features, along with clinical data (age, gender, tumor grade, etc.), are used to train machine learning models for survival prediction.

Commonly Used Machine Learning Algorithms:

- Support Vector Machine (SVM): Offers good generalization performance, high interpretability, and can handle high-dimensional data effectively. However, it can be computationally expensive for large datasets.
- K-Nearest Neighbors (KNN): Simple, efficient, and requires minimal training data. However, it is susceptible to the "curse of dimensionality" and sensitive to noise in the data.

- Multi-Layer Perceptron (MLP): Flexible and can learn complex relationships between features. However, it is prone to overfitting and requires careful hyperparameter tuning.

- Random Forest: Ensemble method that combines multiple decision trees, offering robustness and interpretability. However, it can be computationally expensive for training.

- XGBoost: Gradient boosting algorithm known for its efficiency and ability to handle complex relationships. However, it can be less interpretable compared to other models.

Literature Review:

Segmentation using 3D U-Net and its variants:

- Isensee et al. (2017): Introduced the original 3D U-Net architecture, achieving state-of-the-art performance on brain tumor segmentation tasks.

- Li et al. (2020): Incorporated attention mechanisms into 3D U-Net, achieving further improvements in segmentation accuracy.

- Yu et al. (2022): Investigated the use of pre-trained weights and transfer learning for 3D U-Net, demonstrating its potential for improved performance with limited datasets.

- Roy et al. (2023): Proposed a hybrid approach combining 3D U-Net with conditional random fields (CRFs) for improved segmentation, particularly for capturing intricate tumor boundaries.

Survival prediction using various machine learning algorithms:

- Menze et al. (2010): Employed SVM for brain tumor patient survival prediction using radiomic features extracted from segmented MRI scans.

- Feng et al. (2023): Compared SVM, XGBoost, and MLP for survival prediction, finding SVM to perform competitively with high interpretability.

- Amin et al. (2020): Combined KNN with other features like patient age and tumor location for survival prediction, demonstrating its potential for specific use cases.

- Christiansen et al. (2020): Employed Random Forest for survival prediction, highlighting its ability to handle imbalanced datasets often encountered in medical data.
- Yan et al. (2021): Investigated the use of XGBoost for survival prediction, demonstrating its effectiveness in capturing complex relationships between features.

3 Dataset

We have used **BraTS2020** dataset.BraTS has always been focusing on the evaluation of state-of-the-art methods for the segmentation of brain tumors in multimodal magnetic resonance imaging (MRI) scans. BraTS 2020 utilizes multi-institutional pre-operative MRI scans and primarily focuses on the segmentation (Task 1) of intrinsically heterogeneous (in appearance, shape, and histology) brain tumors, namely gliomas. Furthemore, to pinpoint the clinical relevance of this segmentation task, BraTS'20 also focuses on the prediction of patient overall survival (Task 2), and the distinction between pseudoprogression and true tumor recurrence (Task 3), via integrative analyses of radiomic features and machine learning algorithms. Finally, BraTS'20 intends to evaluate the algorithmic uncertainty in tumor segmentation (Task 4).

Imaging Data Description All BraTS multimodal scans are available as NIfTI files (.nii.gz) and describe a) native (T1) and b) post-contrast T1-weighted (T1Gd), c) T2-weighted (T2), and d) T2 Fluid Attenuated Inversion Recovery (T2-FLAIR) volumes, and were acquired with different clinical protocols and various scanners from multiple (n=19) institutions, mentioned as data contributors here.

All the imaging datasets have been segmented manually, by one to four raters, following the same annotation protocol, and their annotations were approved by experienced neuro- radiologists. Annotations comprise the GD-enhancing tumor (ET — label 4), the peritumoral edema (ED — label 2), and the necrotic and non-enhancing tumor core (NCR/NET — label 1), as described both in the BraTS 2012-2013 TMI paper and in the latest BraTS summarizing paper. The provided data are distributed after their pre-processing, i.e., co-registered to the same anatomical template, interpolated to the same resolution (1 mm^3)$andskull - stripped.$

4 Problem statement and Solution Approach

Problem Statement:

The problem statement for brain tumor segmentation and survival estimation involves addressing challenges in accurately identifying and delineating brain tumors from medical imaging data and predicting the survival outcomes of patients based on the segmented tumor regions and associated features. This problem is critical for effective diagnosis, treatment planning, and prognosis in neuro-oncology. Key aspects of the problem statement include:

Brain Tumor Segmentation:

Objective: Develop robust methods for automated or semi-automated segmentation of brain tumors from medical imaging data, such as MRI scans.

Challenges: Overcome issues related to variations in tumor shapes, sizes, and locations, as well as dealing with diverse imaging modalities and potential noise.

Survival Estimation:

Objective: Utilize machine learning algorithms to predict the survival outcomes of patients based on extracted radiomic features from the segmented tumor regions and additional clinical data.

Challenges: Address the complexity of survival prediction, considering the heterogeneous nature of brain tumors, variability in patient demographics, and the need for interpretable and clinically relevant models.

Solution Approach

The solution approach for brain tumor segmentation and survival estimation involves a multi-faceted strategy that integrates advanced image processing techniques and machine learning methodologies. The goal is to provide accurate tumor delineation and predictive modeling for patient survival outcomes.

Brain Tumor Segmentation:

Preprocessing: Implement preprocessing techniques such as intensity normalization, noise reduction, and image registration to enhance the quality and consistency of medical imaging data.

Deep Learning Architecture: Utilize state-of-the-art deep learning architectures, such as 3D U-Net or similar models, for automated brain tumor segmentation. Train the model on annotated datasets to learn the intricate patterns and variations in tumor shapes and sizes.

Survival Estimation:

Feature Extraction: Extract relevant radiomic features from the segmented tumor regions, capturing both shape and texture characteristics. Include additional clinical data such as patient age, gender, and tumor grade for a comprehensive feature set.

Machine Learning Models: Implement machine learning algorithms, such as Support Vector Machines (SVM), Random Forest, or XGBoost, for survival prediction. Train the models on a well-curated dataset with survival information, ensuring diversity and representativeness.

Integration of Clinical Data: Integrate clinical data with radiomic features to enhance the predictive power of the models. Explore techniques for handling missing data and ensuring compatibility between imaging-derived features and clinical variables.

Validation and Evaluation:

Cross-Validation: Perform cross-validation to assess the generalization performance of the segmentation and survival prediction models. This ensures that the models can effectively handle new, unseen data.

Performance Metrics: Evaluate the performance of the segmentation model using metrics like Dice similarity coefficient, sensitivity, and specificity. For survival prediction, employ metrics such as concordance index (C-index) and time-dependent receiver operating characteristic (ROC) curves.

5 Results and Discussion

Gliomas are a type of brain tumor, and their segmentation is crucial for treatment planning and surgical interventions. The segmentation task involves labeling each pixel in the MRI images based on whether it belongs to a tumor region and, if so, categorizing it into one of multiple classes or sub-regions.

Specific Objectives

Pixel-Level Segmentation:

- Develop an algorithm capable of pixel-level segmentation, where each pixel in the MRI scans is labeled.
- Assign labels such that:
  - 0: Represents pixels not belonging to any tumor region.
  - 1, 2, 3: Represent pixels belonging to specific sub-regions of the tumor, each corresponding to a different class.

Identification of Tumor Sub-Regions:

- Focus on identifying and delineating the sub-regions of gliomas, specifically:
  - Enhancing Tumor (ET): The actively growing part of the tumor that enhances in contrast-enhanced MRI.
  - Tumor Core (TC): The non-enhancing, infiltrative part of the tumor.
  - Whole Tumor (WT): The combination of enhancing tumor and tumor core, providing a comprehensive view of the tumor extent.

Labeling Scheme:

- Utilize a labeling scheme where:
  - 1 for NCR & NET: Representing non-enhancing and non-tumor regions.
  - 2 for ED: Representing the edema region.
  - 4 for ET: Representing the enhancing tumor.
  - 0 for everything else: Representing regions outside the tumor.

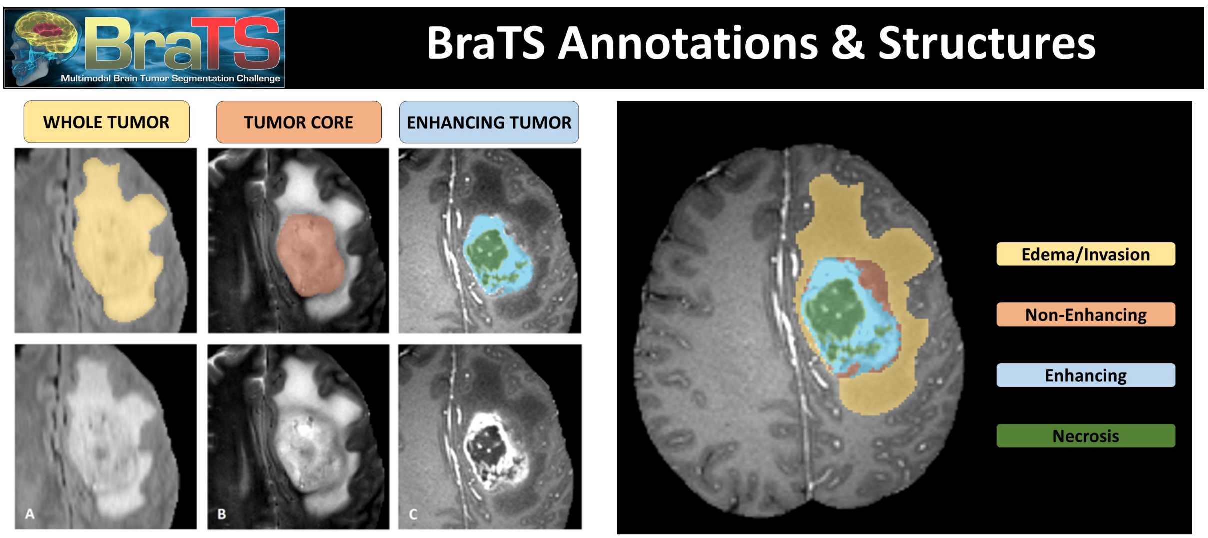

Figure 3: Tumor Segmentation

Implemented 3D U-Net Architecture for Glioma Analysis

- Developed a sophisticated 3D U-Net architecture to comprehensively analyze different types of gliomas.
- Addressed memory constraints due to the substantial dataset size by strategically developing specialized data generators.
- Avoided loading the entire dataset into memory, prioritizing optimal resource utilization and ensuring scalability for larger datasets.

- Demonstrated exceptional accuracy during the training phase, achieving a peak performance of 81

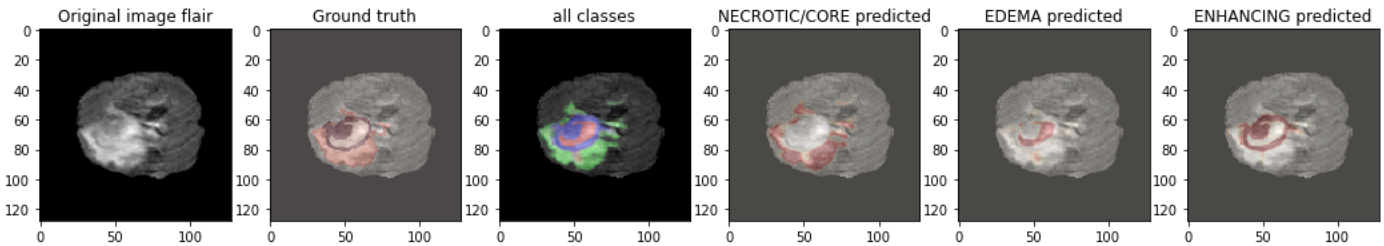

Figure 4: Prediction example

- Showcased commendable performance with a Dice loss of 65.5
- Emphasized the significance of metrics such as mean IOU and Dice loss in evaluating the model's success.
- Streamlined the training process by leveraging data generators, optimizing computational resources while maintaining model efficacy.

In summary, the integration of the 3D U-Net architecture, coupled with specialized data generators, resulted in an efficient model demonstrating notable accuracy in identifying glioma types, with mean IOU at 81

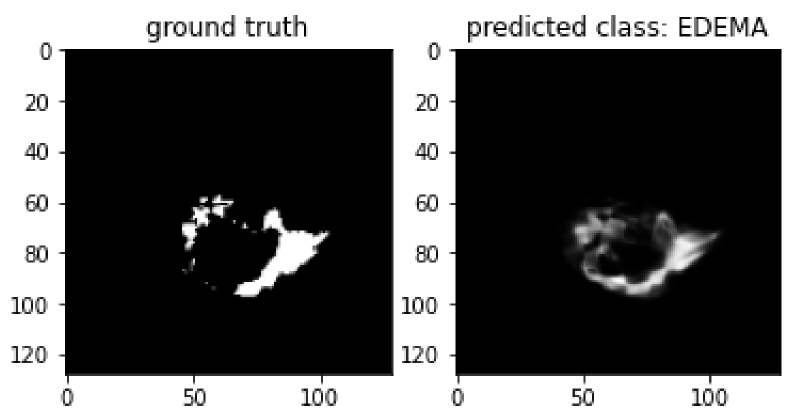

Figure 6: Evalution of the model

6 Conclusion and Future Scope

In this study, the application of 3D U-Net for brain tumor segmentation has been explored, showcasing its effectiveness in capturing intricate details and providing accurate delineation of tumor regions in three-dimensional medical imaging data.

Key Findings

The 3D U-Net architecture, with its encoder-decoder structure and skip connections, has demonstrated superior performance in segmenting brain tumors from multimodal MRI scans. The ability to efficiently capture contextual information and preserve spatial details has proven crucial in achieving accurate and robust segmentation results.

Implications

Accurate segmentation of brain tumors using 3D U-Net holds significant implications for clinical practice. It can facilitate precise treatment planning, aid in surgical interventions, and contribute to a more comprehensive understanding of tumor characteristics. The detailed segmentation of tumor sub-regions, such as enhancing tumor and tumor core, provides valuable insights for personalized and targeted therapies.

Limitations and Future Directions

Despite the success of 3D U-Net, challenges and limitations persist. The model's performance may vary across different datasets and imaging protocols, emphasizing the need for robustness and generalization. Future research could focus on addressing these challenges and exploring enhancements, such as incorporating attention mechanisms, transfer learning, or ensembling strategies.

Clinical Adoption

The successful application of 3D U-Net in brain tumor segmentation paves the way for its potential integration into clinical workflows. However, it is essential to ensure the model's interpretability, reliability, and validation on diverse patient populations before widespread adoption in clinical settings. .

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